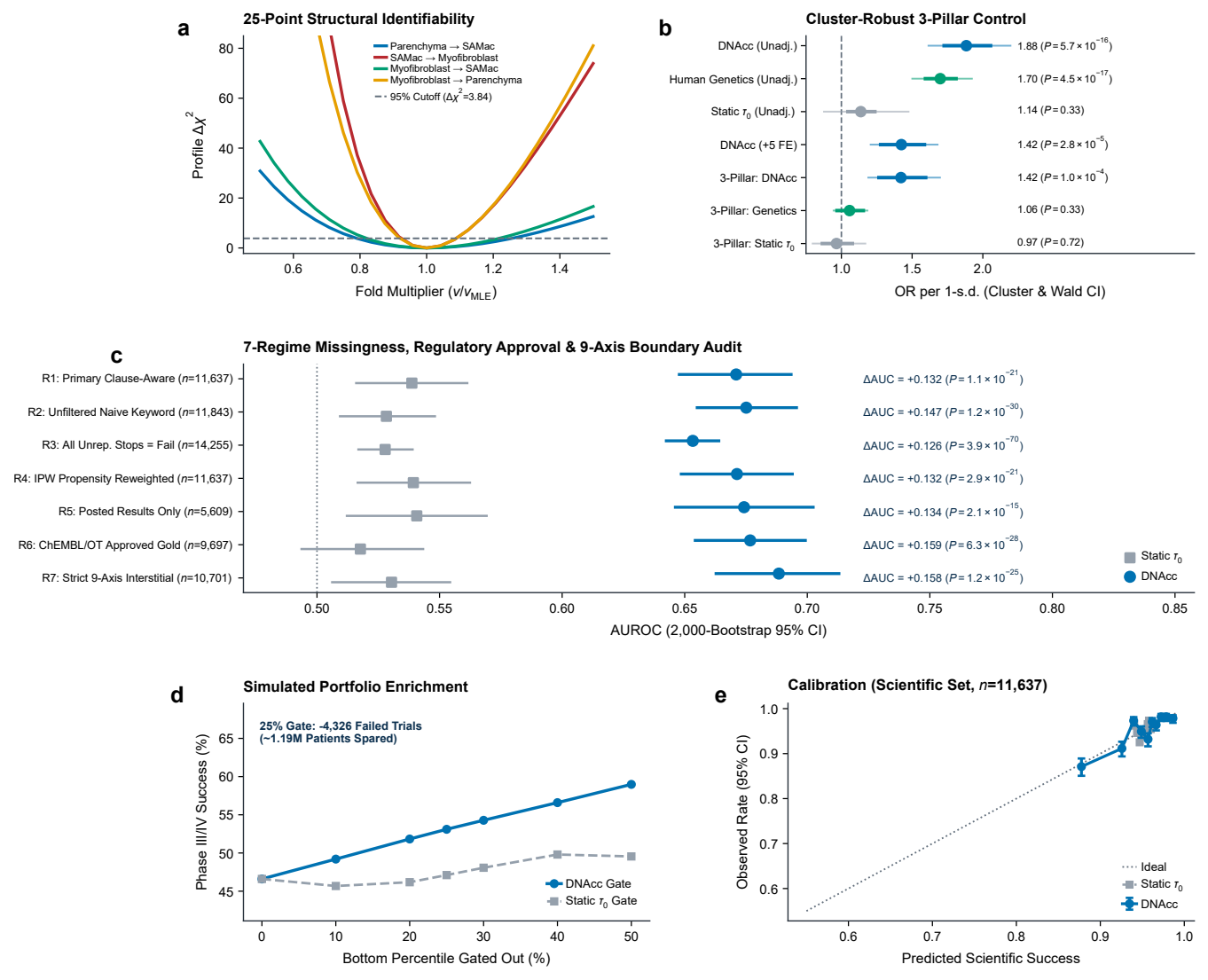

a
25-Point Structural Identifiability
Parenchyma → SAMac
SAMac → Myofibroblast
Myofibroblast → SAMac
Myofibroblast → Parenchyma
95% Cutoff ($\Delta\chi^2$=3.84)
Profile $\Delta\chi^2$
Fold Multiplier (v/v_{MLE})
b
Cluster-Robust 3-Pillar Control
DNAcc (Unadj.)
Human Genetics (Unadj.)
Static τ_0 (Unadj.)
DNAcc (+5 FE)
3-Pillar: DNAcc
3-Pillar: Genetics
3-Pillar: Static τ_0
1.88 ($P = 5.7 \times 10^{-16}$)
1.70 ($P = 4.5 \times 10^{-17}$)
1.14 ($P = 0.33$)
1.42 ($P = 2.8 \times 10^{-5}$)
1.42 ($P = 1.0 \times 10^{-4}$)
1.06 ($P = 0.33$)
0.97 ($P = 0.72$)
OR per 1-s.d. (Cluster & Wald CI)
c
7-Regime Missingness, Regulatory Approval & 9-Axis Boundary Audit
R1: Primary Clause-Aware (n=11,637)
R2: Unfiltered Naive Keyword (n=11,843)
R3: All Unrep. Stops = Fail (n=14,255)
R4: IPW Propensity Reweighted (n=11,637)
R5: Posted Results Only (n=5,609)
R6: ChEMBL/OT Approved Gold (n=9,697)
R7: Strict 9-Axis Interstitial (n=10,701)
ΔAUC = +0.132 ($P = 1.1 \times 10^{-21}$)
ΔAUC = +0.147 ($P = 1.2 \times 10^{-30}$)
ΔAUC = +0.126 ($P = 3.9 \times 10^{-70}$)
ΔAUC = +0.132 ($P = 2.9 \times 10^{-21}$)
ΔAUC = +0.134 ($P = 2.1 \times 10^{-15}$)
ΔAUC = +0.159 ($P = 6.3 \times 10^{-28}$)
ΔAUC = +0.158 ($P = 1.2 \times 10^{-25}$)
Static τ_0
DNAcc
AUROC (2,000-Bootstrap 95% CI)
d
Simulated Portfolio Enrichment
25% Gate: -4,326 Failed Trials
(~1.19M Patients Spared)
Phase III/IV Success (%)
DNAcc Gate
Static τ_0 Gate
Bottom Percentile Gated Out (%)
e
Calibration (Scientific Set, n=11,637)
Observed Rate (95% CI)
Ideal
Static τ_0
DNAcc
Predicted Scientific Success